

Detailed anatomy-aware JEPA pipeline

A · Prepare each clip independently

Model input

Short-gap filling; pelvis centering;
body scaling; 64-frame resizing.

X: $B \times 64 \times 33 \times 3$

Validity: $B \times 64 \times 33$

Signed movement contrast

Original timestamps and valid pairs.

No filling or temporal resizing.

Five bilateral pairs define target y.

Image-derived coordinates.

B · Fix five source-video folds

625 clips / 93 sources; every clip and derived view inherits its source fold.

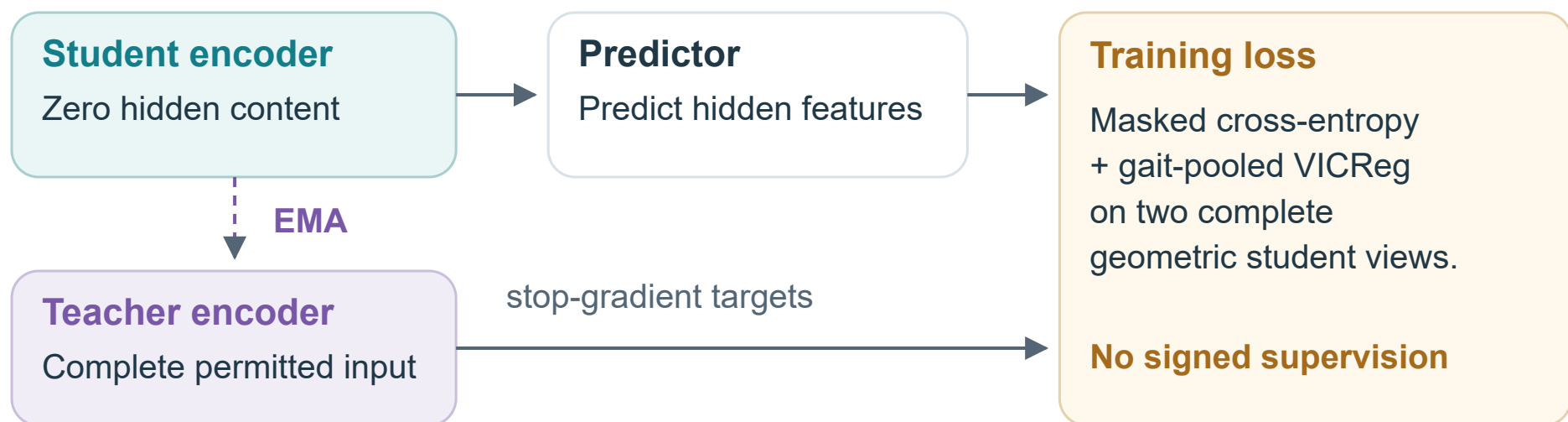
C · Keep the full anatomical grid; vary hidden targets

Patches: $B \times 16 \times 33 \times 12 \rightarrow$ tokens: $B \times 528 \times 96$.

12: gait vs all joints. 15–18: motion or connected-region masks.

All 33 joints stay in the input. Valid and hidden masks: $B \times 16 \times 33$.

D · Train only on outer-training sources



E · Notebook 09 adds an explicit encoder relation

$M(X)[t, j] = (-x, y, z)[t, \pi(j)]$; compare $Z(MX)$ with $P Z(X)$.

Aligned valid tokens; identity channel action; detached energy denominator.

SYNTHETIC ONLY · 16 frames / 16 channels · two extra encoder passes

F · Freeze the encoder, then fit and evaluate the readout

Fit scaling and ridge only on training-source groups; predict held-out videos.

Compare with initial S-JEPA weights using the same pose extractor and schema.

Pool held-out folds per seed; weight every source video equally.

Video-bootstrap intervals condition on fitted models; seeds reuse the cohort.

Preparation precedes fold-file creation; training sampling follows source grouping.

Tensor axes and joint identities persist; interpolation and resizing change coordinates.